SIMATS ENGINEERING
Engineer to Excel

SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA1017 SOFTWARE ENGINEERING

INTELLIGENT DISASTER EARLY WARNING AND EVACUATION MANAGEMENT SYSTEM

S. NO.	MEMBER NAME	REGISTER NUMBER
1	RAGUL SG	192525007
2	RAKKESH KARAN R	192524253
3	DAVIS DAMIEN M	192521124
4	GANTA RAM CHARAN	192525043
5	OSURU SIDDHI RISHWANTH	192472148
6	DURGAM HITHESH	192472167
7	GANDLURI VEDA VIKAS GOWD	192472123

1. PROBLEM STATEMENT AND PROBLEM FORMULATION

Disaster management involves information from IoT sensors, weather stations, seismic systems, environmental devices, satellite sources, GIS platforms, emergency agencies, and citizens. When these sources are disconnected, authorities may receive delayed or incomplete information. This can lead to late warnings, unsafe evacuation decisions, poor shelter-resource management, and weak coordination among government authorities, police, fire and rescue services, medical teams, and relief organizations.

The proposed project develops a full-stack, containerized **Intelligent Disaster Early Warning and Evacuation Management System** to integrate these activities into one platform. The system provides live monitoring, disaster-risk assessment, affected-area mapping, location-specific alerts, evacuation-route recommendations, shelter management, citizen incident reporting, and emergency-response coordination.

The MVP uses simulated sensor and environmental data with a transparent rule-based risk assessment. It classifies conditions into Normal, Watch, Warning, and Critical levels. Future versions may use trained AI/ML models with reliable historical datasets. GIS tools display hazard zones, shelters, emergency facilities, road closures, and evacuation routes. Authorities can monitor shelter occupancy, supplies, response teams, incidents, and evacuation progress through a centralized dashboard. Citizens can receive safety information, locate shelters, view routes, and report emergencies.

The project follows the **Agile Scrum methodology**. Requirements are collected from citizens, disaster-management authorities, emergency responders, and administrators. Features are organized in a product backlog and prioritized using the **MoSCoW technique**. Development is divided into sprints with defined goals, tasks, acceptance criteria, testing, and reviews. A Scrum board tracks work from Backlog to Done.

2. OBJECTIVES AND EXPECTED OUTCOMES

The main objective is to develop a simple, integrated web platform that demonstrates how data, risk information, communication, evacuation planning, shelter operations, and citizen participation can be connected within one emergency-management workflow.

Objective	**Expected outcome**	**Prototype evidence**
Collect and display multi-source monitoring data	Authorities can inspect simulated sensor readings and source health	Live Monitoring page and sensor table
Estimate and communicate potential risk	Users can understand hazard type, severity, confidence, and contributing signals	Risk cards, charts, and model-assessment panel
Provide targeted multi-channel alert workflows	An authority can compose, preview, validate, and record a demo alert	Alert Composer and Alerts Center
Support GIS-based risk understanding	Users can view hazard zones, sensors, shelters, facilities, and routes	Risk Map page and layer controls
Recommend evacuation options	Users can compare simulated routes using time, distance, safety, accessibility, and closures	Evacuation Planner
Monitor shelter resources	Authorities can inspect capacity, occupancy, facilities, supplies, and shortages	Shelter Management page
Enable citizen reporting	Citizens can submit simulated	Citizen Portal and Incidents

	incident reports that authorities can triage	page
Improve agency coordination	Teams can be viewed, assigned, and tracked through local workflow state	Response Coordination page

3. REQUIREMENTS, CONSTRAINTS, AND ASSUMPTIONS

3.1 FUNCTIONAL REQUIREMENTS

ID	Requirement
FR-01	The system shall display an operational overview of active threats, population at risk, shelter capacity, evacuation progress, response teams, and unresolved incidents.
FR-02	The system shall display simulated sensor readings, source status, thresholds, and historical trends.
FR-03	The system shall classify potential risk into normal, watch, warning, and critical levels using transparent demo logic.
FR-04	The system shall display GIS layers for hazard zones, sensors, shelters, emergency facilities, and evacuation routes.
FR-05	An authority user shall be able to create, preview, validate, and record a simulated targeted alert.
FR-06	The system shall provide a simulated evacuation route recommendation using user-selected constraints.
FR-07	The system shall display shelter occupancy, capacity, accessibility, facilities, and supplies.
FR-08	Citizens shall be able to submit incident reports with category, location, description, and contact preference.

FR-09	Authorities shall be able to filter, triage, assign, update, and resolve incident reports.
FR-10	Authorities shall be able to view response teams and assign a team to a priority incident.
FR-11	The application shall support local persistence and a reset-demo-data function.
FR-12	The system shall provide a citizen-facing portal that does not expose authority-only operational information.

3.2 NON-FUNCTIONAL REQUIREMENTS

The platform should be responsive, readable, keyboard accessible, and usable at mobile, tablet, and desktop widths. It should provide visible focus states, clear error handling, meaningful empty states, and status communication that does not depend on colour alone. It should render efficiently with a small local dataset and should remain usable when the map service is unavailable by providing a labelled fallback map view.

The application should be maintainable through typed interfaces, reusable components, separation of mock data from UI logic, and service abstractions that can later be replaced by API calls. It should be honest about the prototype's limitations and must not imply that simulated data is live or that route recommendations are official instructions.

3.3 CONSTRAINTS

The MVP is constrained by the lack of authenticated IoT streams, official hazard feeds, a production database, validated AI/ML training data, live SMS/email credentials, authoritative road-closure data, and real shelter-management integrations. It is also constrained by the need to keep the first implementation simple enough for an academic demonstration.

3.4 ASSUMPTIONS

The prototype assumes a fictional monitoring region called Riverdale County, deterministic mock records, a single-browser demonstration session, a trusted authority operator, and manually verified alerts. The terms "AI assessment," "risk," "route," "shelter status," and "population at risk" refer to simulated or calculated prototype values unless explicitly marked as real and externally verified.

4. APPLICATION OF RELEVANT COURSE KNOWLEDGE AND CONCEPTS

The final mapping must be aligned with concepts actually covered in the team's course. The following table is a proposed mapping and should be edited before submission.

Course concept	Application in this project	Evidence
Software engineering	Requirements analysis, modular design, separation of concerns, testing, maintainability	Requirements matrix, component structure, test cases
Web development	React pages, routing, forms, responsive layout, reusable components	Source code and execution screenshots
Algorithms	Risk classification, alert prioritization, route scoring, shelter recommendation	Pseudocode and flowchart
AI/ML	Model-assessment concept, confidence display, future ML service boundary, responsible-AI limitation	Risk panel and future-work analysis
GIS and spatial reasoning	Hazard zones, map layers, exposure, shelters, facilities, and routes	Risk Map page and architecture diagram
Data management	Typed records, filtering, local persistence, normalization, and future database boundary	Data model and mock-data module
Human-computer interaction	Role-based views, clear severity hierarchy, citizen-first actions, accessibility	UI screenshots and usability observations
Data visualization	KPI cards, charts, status distributions, occupancy bars, and activity timelines	Dashboard and monitoring screenshots
Cybersecurity	Role separation, privacy note,	Requirements and limitations

	secure-authentication future boundary, no real personal data	
Cloud computing	Scalable storage and processing as a proposed production extension	Architecture and future integration section

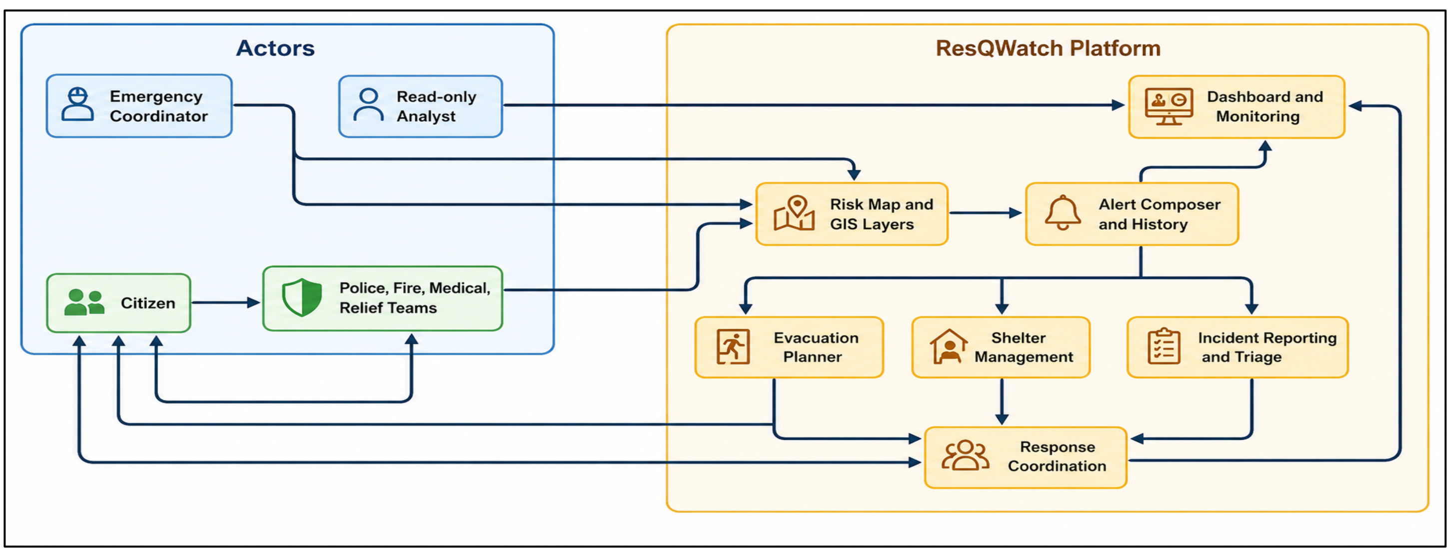
5. DESIGN, PROPOSED SOLUTION, AND METHODOLOGY

5.1 PROPOSED SOLUTION

ResQWatch is designed as a modular web application with an authority operations console and a citizen portal. The authority console provides a persistent navigation shell containing overview, monitoring, risk map, alerts, evacuation, shelters, incidents, teams, and settings. The citizen portal reduces complexity and focuses on active local risk, practical guidance, nearby shelters, evacuation lookup, and incident reporting.

The platform is organised around four operational stages: observe, assess, communicate, and respond. Observation receives simulated measurements and source-health information. Assessment applies transparent demo rules to hazard indicators and exposure data. Communication prepares a targeted message for authority review and records the configured channels in demo mode. Response connects route planning, shelter capacity, incident triage, and team coordination.

5.2 SYSTEM ARCHITECTURE

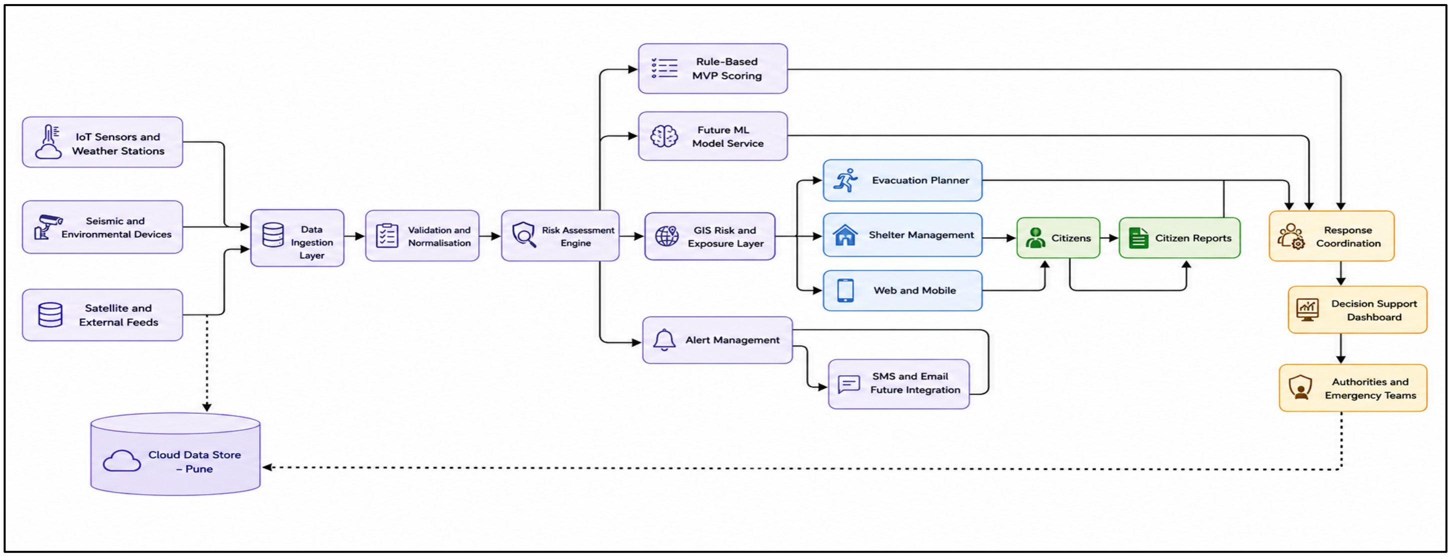


System architecture

The architecture separates input data, validation, assessment, GIS presentation, alert management, evacuation planning, shelter management, and response coordination. A future production version could replace the local mock-data repository with APIs, streaming ingestion, a geospatial database, model-serving infrastructure, and secure notification services.

5.3 USER ROLES

The **Emergency Coordinator** can review risk, create demo alerts, manage shelters, triage incidents, and assign teams. The **Read-only Analyst** can review dashboards, maps, and trends but cannot issue or modify operational actions. **Response teams** can receive assignments and update activity status. **Citizens** can view public guidance, find shelters, view route recommendations, and submit incident reports.



User roles and platform flow

5.4 DEVELOPMENT METHODOLOGY

An iterative, risk-first development methodology is appropriate for the project. The team should first establish the application shell and design system, then implement the dashboard and shared data structures, followed by monitoring, maps, alerts, evacuation, shelters, incidents, teams, and citizen views.

6. ALGORITHMS, PSEUDOCODE, AND FLOWCHART

6.1 RISK CLASSIFICATION

```
FUNCTION classifyRisk(hazardIntensity, exposure, vulnerability, dataQuality):
    IF dataQuality is stale OR dataQuality is invalid:
        RETURN {level: "WATCH", confidence: "LOW", reason: "Verification required"}

    score = 0.45 * hazardIntensity
          + 0.25 * exposure
          + 0.20 * vulnerability
```

```
        + 0.10 * dataQuality

    IF score >= 0.80:
        level = "CRITICAL"
    ELSE IF score >= 0.60:
        level = "WARNING"
    ELSE IF score >= 0.35:
        level = "WATCH"
    ELSE:
        level = "NORMAL"

    confidence = estimateConfidence(dataQuality, sourceAgreement)
    RETURN {level, score, confidence}
END FUNCTION
```

6.2 ALERT PRIORITIZATION

```
FUNCTION prioritizeAlert(riskLevel, populationExposed, timeToImpact, vulnerability):
    severityWeight = weightFor(riskLevel)
    urgencyWeight = inverseScale(timeToImpact)
    priority = 0.40 * severityWeight
             + 0.25 * normalize(populationExposed)
             + 0.20 * urgencyWeight
             + 0.15 * vulnerability
    RETURN sortDescendingByPriority(priority)
END FUNCTION
```

6.3 SAFEST-ROUTE SELECTION

```
FUNCTION selectRoute(routes, constraints):
    feasibleRoutes = []
    FOR route IN routes:
        IF route.hasClosedRoad AND constraints.avoidClosedRoads:
            CONTINUE
        IF route.entersHazardZone AND constraints.avoidHazardZones:
            CONTINUE
        IF constraints.accessibleOnly AND NOT route.isAccessible:
            CONTINUE
        feasibleRoutes.append(route)

    IF feasibleRoutes is empty:
        RETURN {status: "UNAVAILABLE", message: "No simulated route satisfies constraints"}
```

```
    FOR route IN feasibleRoutes:
        route.score = 0.35 * route.safetyScore
                    + 0.25 * inverseScale(route.travelTime)
                    + 0.20 * inverseScale(route.distance)
                    + 0.10 * route.accessibilityScore
                    + 0.10 * route.reliabilityScore
    RETURN route with highest score
END FUNCTION
```

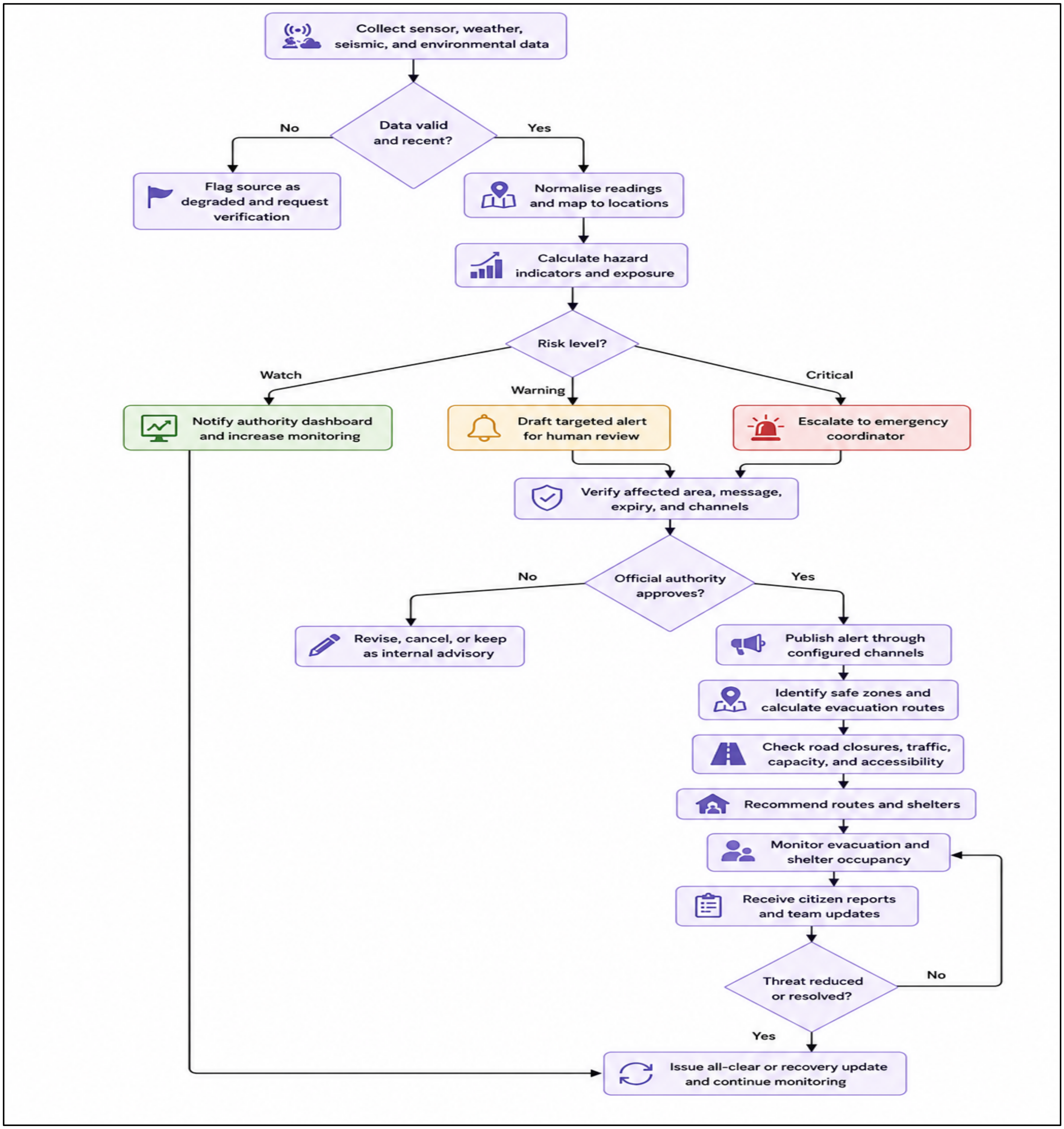
6.4 SHELTER RECOMMENDATION

```
FUNCTION recommendShelter(shelters, requirements):
    candidates = shelters WHERE status = OPEN
    candidates = candidates WHERE availableCapacity >= requirements.people
    IF requirements.accessible:
        candidates = candidates WHERE accessible = TRUE
    IF requirements.medicalSupport:
        candidates = candidates WHERE medicalSupport = TRUE

    FOR shelter IN candidates:
        shelter.score = 0.35 * capacityAvailability(shelter)
                      + 0.25 * inverseScale(shelter.distance)
                      + 0.20 * facilityMatch(shelter, requirements)
                      + 0.20 * supplyReadiness(shelter)
    RETURN top three candidates sorted by score
END FUNCTION
```

6.5 INCIDENT PRIORITIZATION

```
FUNCTION prioritizeIncident(incident):
    categoryWeight = weightForCategory(incident.category)
    severityWeight = weightFor(incident.severity)
    vulnerabilityWeight = incident.involvesVulnerablePerson ? 1.0 : 0.0
    ageWeight = increaseWithElapsedTime(incident.createdAt)
    incident.priority = 0.40 * severityWeight
                      + 0.25 * categoryWeight
                      + 0.20 * vulnerabilityWeight
                      + 0.15 * ageWeight
    RETURN incident
END FUNCTION
```



Warning and evacuation flow

The operational flow is intentionally designed so that automated assessment produces a recommendation, while alert publication requires authority verification. This supports responsible use of AI-assisted decision support and avoids representing an unverified model output as an official instruction.

7. IMPLEMENTATION, SOURCE CODE, ENVIRONMENT, AND TOOLS

7.1 PROPOSED ENVIRONMENT

Area	Proposed tool or technology
Frontend	React and TypeScript
Styling	Tailwind CSS and shadcn/ui
Routing	Wouter or existing scaffold router
Icons	Lucide React
Charts	Recharts
Mapping	Existing Google Maps `MapView` component where available
State	React state/context and localStorage for MVP
Validation	React Hook Form and Zod where available
Data	Deterministic fictional Riverdale County mock data
Testing	TypeScript checks, production build, route verification, interaction checks
Future backend	REST/WebSocket APIs, database, PostGIS, model service, messaging providers

7.2 SUGGESTED SOURCE STRUCTURE

```
client/src/
  components/
    AppShell.tsx
    SidebarNavigation.tsx
    TopBar.tsx
    MetricCard.tsx
    SeverityBadge.tsx
    MapControls.tsx
    AlertComposer.tsx
    ShelterDetailDrawer.tsx
    IncidentDetailDrawer.tsx
  contexts/
    DemoDataContext.tsx
    ThemeContext.tsx
```

```
hooks/
  useDemoData.ts
  useLocalStorage.ts
lib/
  types.ts
  mockData.ts
  risk.ts
  routing.ts
  formatters.ts
pages/
  Home.tsx
  LiveMonitoring.tsx
  RiskMap.tsx
  Alerts.tsx
  Evacuation.tsx
  Shelters.tsx
  Incidents.tsx
  ResponseTeams.tsx
  Citizen.tsx
  Settings.tsx
```

The implementation should preserve the scaffold’s existing design tokens and should not place large media assets in the public directory. It should include a map fallback, local error handling, reusable dialogs, empty states, and clear demo labels.

8. TEST CASES AND EXPECTED/ACTUAL RESULTS

Because the current assignment workspace did not include an executable source-code project or captured browser evidence, the “Actual result” column must be completed by the team after implementation and execution. It must not be filled with invented claims.

Test ID	Feature	Preconditions	Steps	Expected result	Actual result	Status	Evidence
TC-01	Dashboard load	Application started	Open `/`	Overview loads with KPI cards, map, charts, and recent alerts	To be recorded after execution	Not executed	Screenshot A
TC-02	Navigation	Application started	Open each sidebar	Every route renders without	To be recorded	Not executed	Screenshot index

			route	blank screen	after execution		
TC-03	Sensor filtering	Monitoring page open	Filter by offline or degraded status	Matching sensor records are shown	To be recorded after execution	Not executed	Screenshot B
TC-04	Risk-map layer toggle	Risk Map open	Toggle shelters and hazard layers	Selected layers appear/disappear and legend updates	To be recorded after execution	Not executed	Screenshot C
TC-05	Invalid alert form	Alert Composer open	Submit without title, severity, or zone	Inline validation prevents submission	To be recorded after execution	Not executed	Screenshot D
TC-06	Demo alert creation	Valid alert fields entered	Preview and confirm demo alert	New alert appears in Active or History and toast is shown	To be recorded after execution	Not executed	Screenshot E
TC-07	Route calculation	Evacuation page open	Select origin, shelter, and constraints	Recommended and alternative simulated routes appear	To be recorded after execution	Not executed	Screenshot F
TC-08	No feasible route	Evacuation page open	Select conflicting constraints	Friendly unavailable state and recovery action appear	To be recorded after execution	Not executed	Screenshot G
TC-09	Shelter update	Shelter detail open	Update occupancy or supplies	Values and status update locally	To be recorded after execution	Not executed	Screenshot H
TC-10	Citizen report	Citizen portal open	Submit valid	Report is added and confirmation	To be recorded	Not executed	Screenshot I

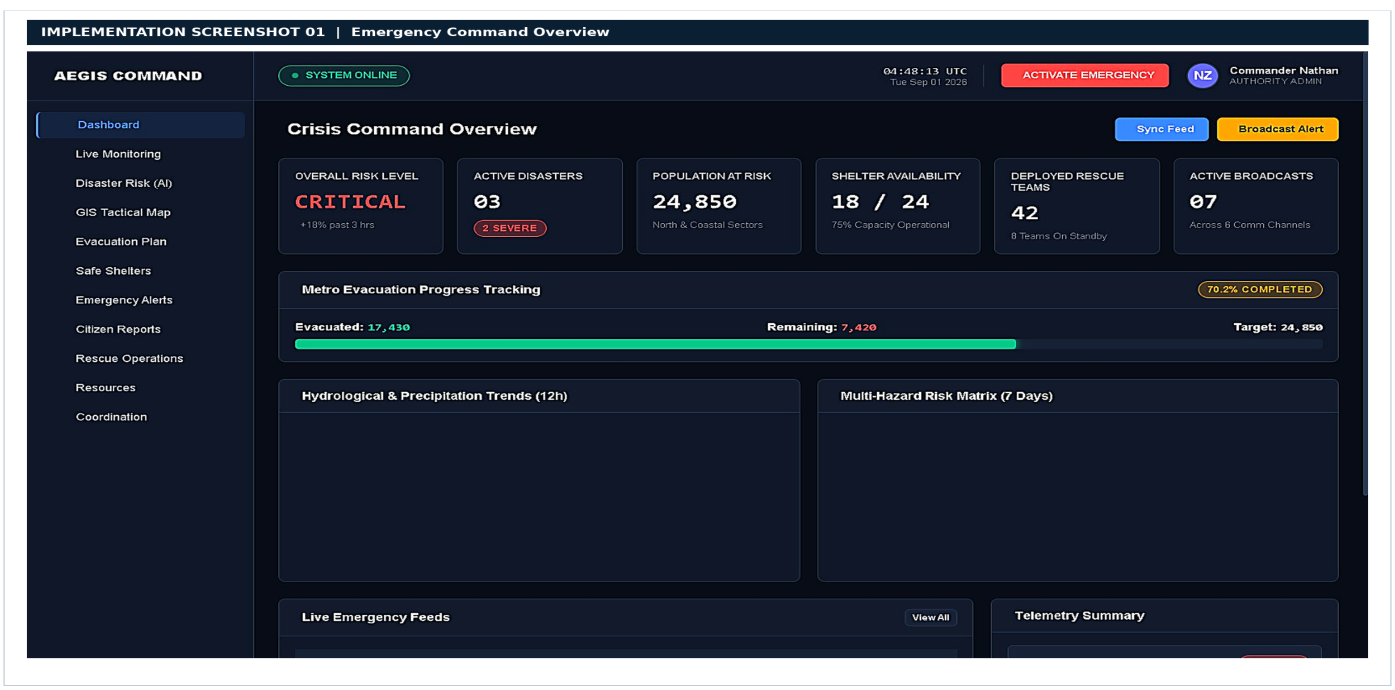
			incident	appears	after execution		
TC-11	Incident triage	Authority incident page open	Assign team and change status	Assignment and activity timeline update	To be recorded after execution	Not executed	Screenshot J
TC-12	Role separation	Citizen role selected	Open authority routes or controls	Authority-only controls are hidden or disabled	To be recorded after execution	Not executed	Screenshot K
TC-13	Map fallback	Map service unavailable	Open map route	Fallback map is displayed with clear label	To be recorded after execution	Not executed	Screenshot L
TC-14	Responsive layout	Browser resized to mobile	Navigate and open dialogs	No important content is clipped; mobile navigation works	To be recorded after execution	Not executed	Mobile screenshots
TC-15	Reset demo data	Demo data modified	Select reset action	Seeded state is restored after confirmation	To be recorded after execution	Not executed	Screenshot M

9. EXECUTION SCREENSHOTS AND OUTPUTS

Figure	**Required output**	**Suggested caption**
Fig. 1	Overview dashboard	Emergency operations overview showing simulated threats, population exposure, shelter capacity, and activity.
Fig. 2	Live monitoring	Simulated sensor readings, source health, environmental trends, and model-assessment panel.
Fig. 3	Risk map	GIS-style hazard zones, sensors, shelters, facilities, and route layers.

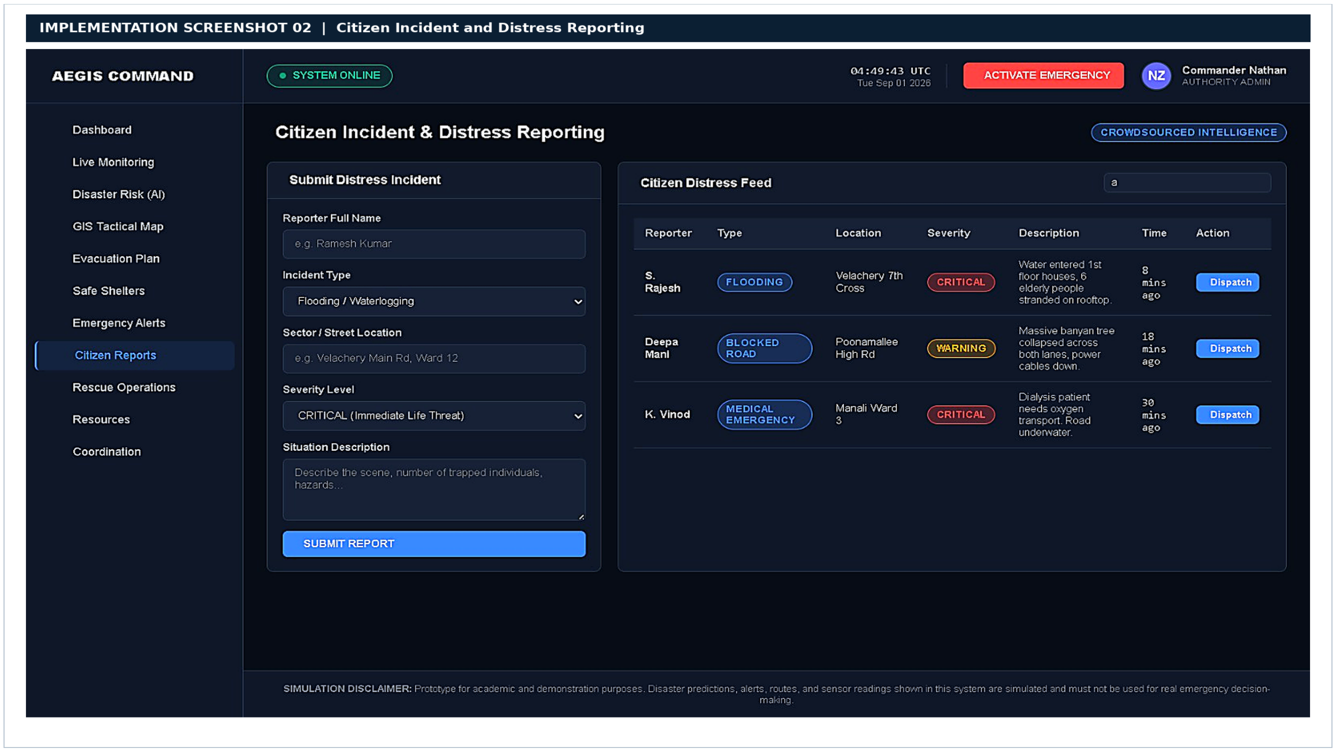
Fig. 4	Alert composer	Multi-step demo alert creation and affected-area selection.
Fig. 5	Alert preview	Desktop or mobile preview showing severity, message, expiry, and channels.
Fig. 6	Evacuation planner	Simulated safest-route comparison with closure and accessibility constraints.
Fig. 7	Shelter management	Capacity, facilities, supplies, and shelter-detail view.
Fig. 8	Citizen reporting	Citizen incident form and validation/confirmation state.
Fig. 9	Response coordination	Team availability, assignment queue, and activity timeline.
Fig. 10	Citizen portal	Public risk status, shelter lookup, route access, and incident reporting.
Fig. 11	Validation state	Example of invalid form feedback, empty state, fallback map, or recovery behavior.

9.1 IMPLEMENTATION SCREENSHOTS FROM UPLOADED ARCHIVE



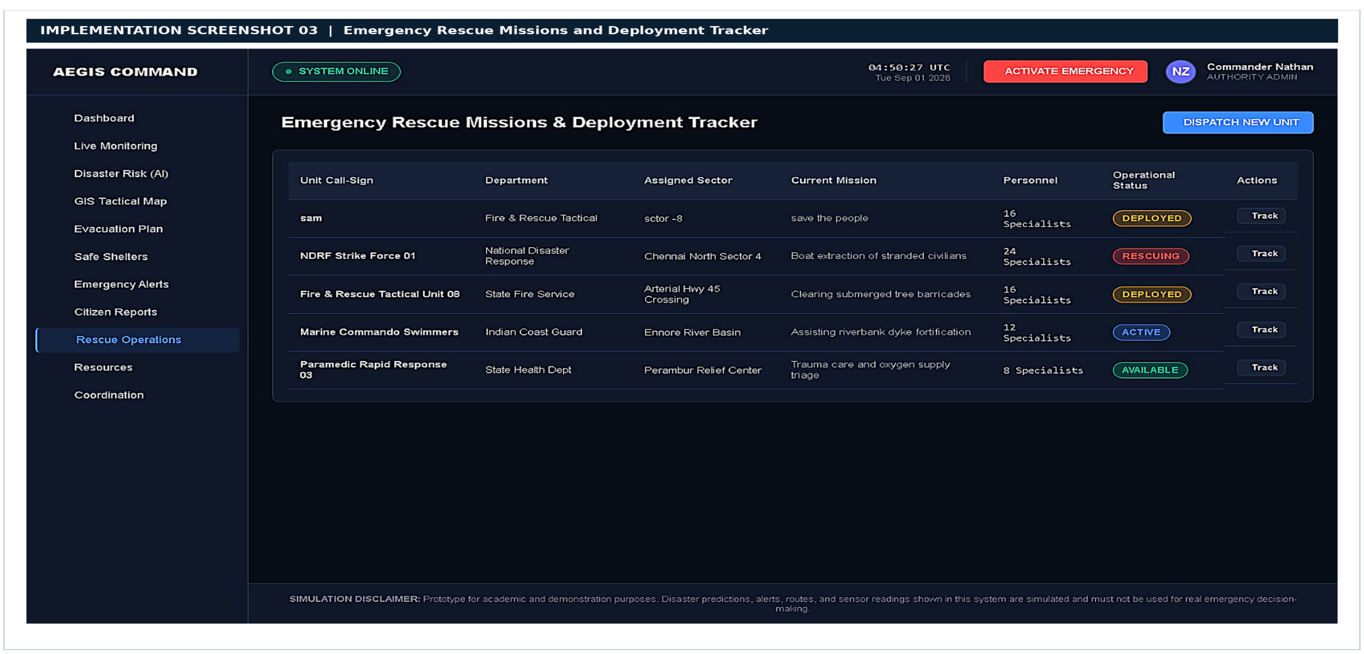
Implementation dashboard

Figure 1. Supplied implementation screenshot of the emergency-operations dashboard showing KPI cards, a risk-map panel, threat levels, recent activity, and demo-mode labelling.



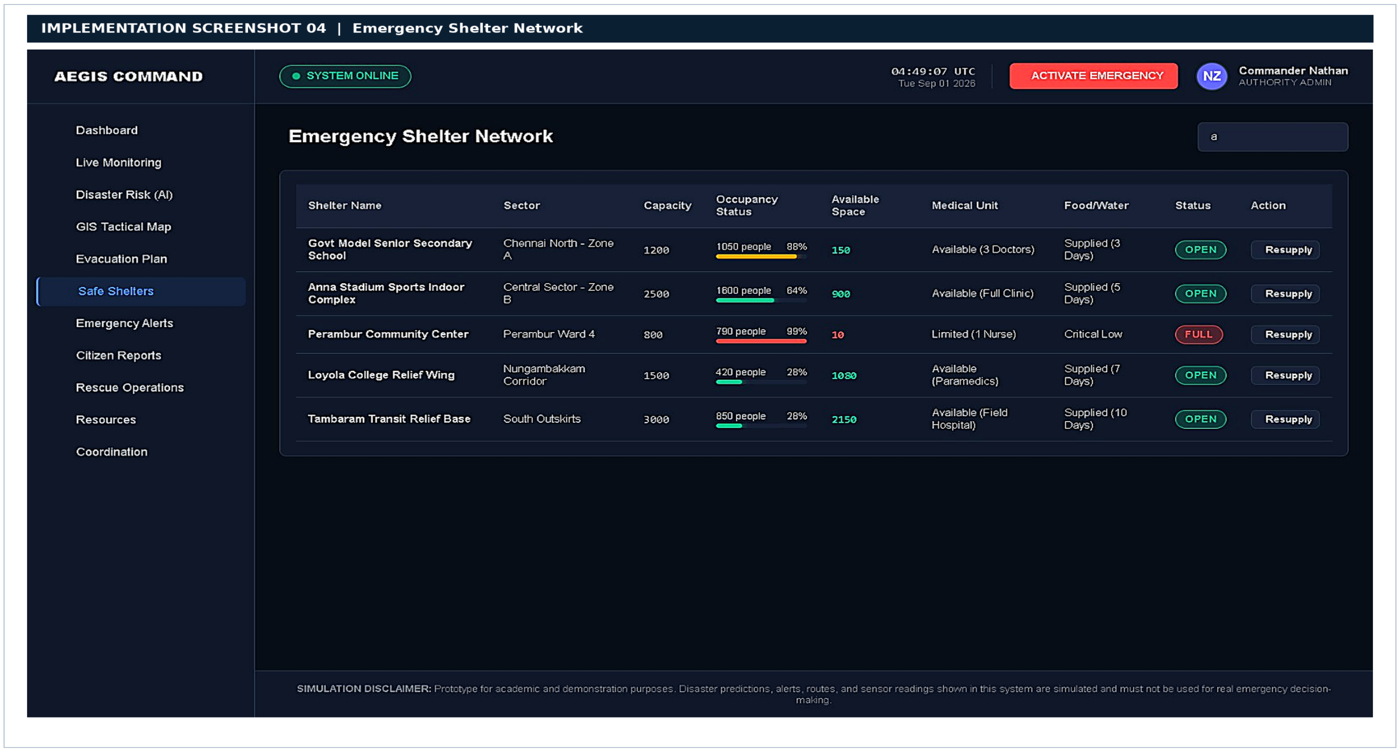
Implementation live monitoring

Figure 2. Supplied implementation screenshot of the live-monitoring page showing sensor health, trends, source status, and a simulated model assessment.



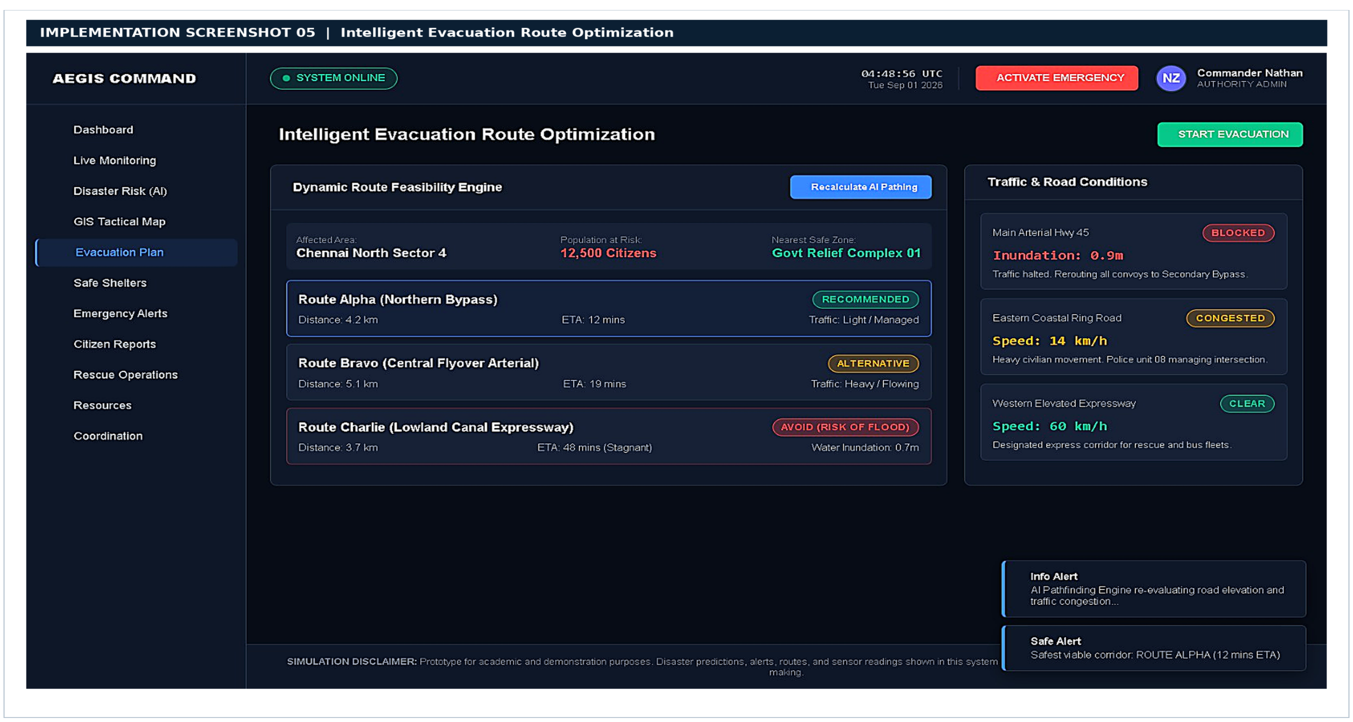
Implementation risk map

Figure 3. Supplied implementation screenshot of the GIS risk map with hazard zones, shelters, facilities, sensors, and evacuation routes.



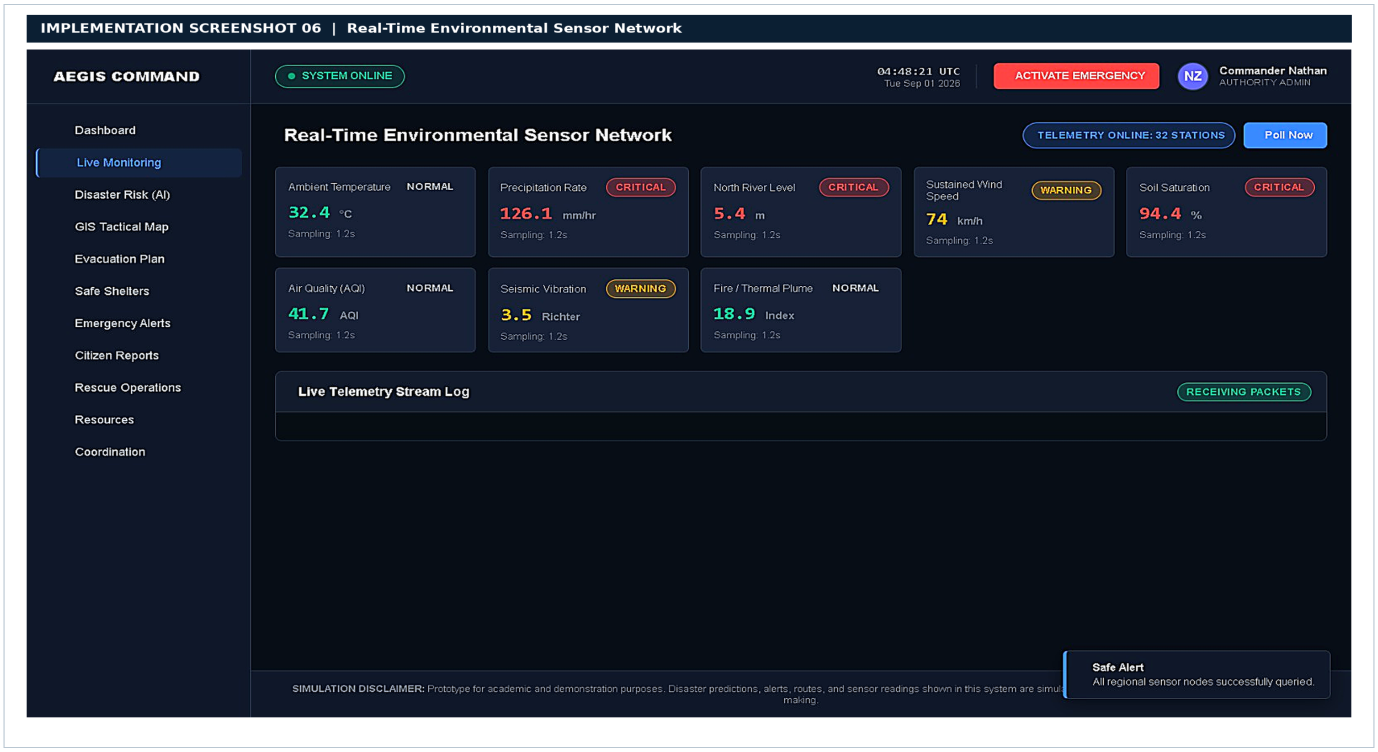
Implementation alert composer

Figure 4. Supplied implementation screenshot of the alert-composer workflow with affected area, message, delivery channels, preview action, and demo-only warning.



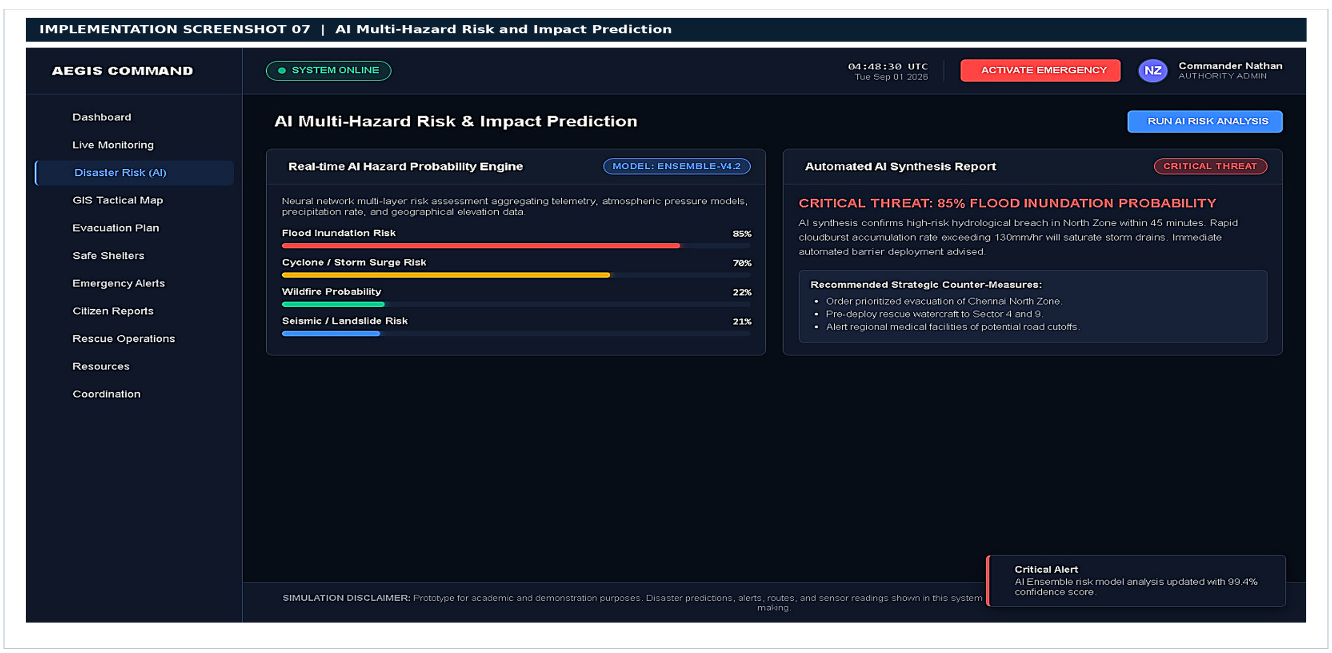
Implementation evacuation planner

Figure 5. Supplied implementation screenshot of the evacuation planner comparing simulated routes using time, score, accessibility, and closure avoidance.



Implementation shelter management

Figure 6. Supplied implementation screenshot of the shelter-management interface showing occupancy, capacity, facilities, status, and supply-related indicators.



Implementation citizen portal

Figure 7. Supplied implementation screenshot of the citizen portal focused on current warning guidance, shelter lookup, route access, and incident reporting.

10. RESULTS AND VALIDATION

The MVP should validate functionality and workflow completeness rather than real-world prediction performance. The following table is a reporting template. Values must be filled using actual observations from the implemented application.

Metric	**Measurement method**	**Result**	**Evidence classification**
Route coverage	Number of required routes that render successfully divided by routes tested	To be measured	Observed
Alert workflow completion	Percentage of valid alert scenarios that pass validation, preview, and local creation	To be measured	Observed
Form validation coverage	Number of required invalid-input cases handled correctly	To be measured	Observed
Responsive route coverage	Number of routes usable at target mobile width	To be measured	Observed
Shelter update consistency	Whether occupancy and supply changes are reflected across relevant views	To be measured	Observed
Map fallback availability	Whether the application remains usable when the map is unavailable	To be measured	Observed
Simulated exposed population	Sum of fictional zone values in selected scenario	Use seeded data	Calculated from mock data
Simulated shelter	Occupancy divided by	Use seeded data	Calculated from mock data

occupancy	capacity for selected shelters		
AI/ML predictive accuracy	Requires labeled historical data and evaluation protocol	Not evaluated in this prototype	Unavailable
Real warning latency	Requires live end-to-end sensor and messaging instrumentation	Not evaluated in this prototype	Unavailable
Casualty or property-damage reduction	Requires operational deployment and controlled comparative study	Not evaluated in this prototype	Unavailable

VALIDATION SCENARIO

The recommended demonstration scenario is a flood-related event in Riverdale County. Heavy rainfall causes elevated river readings, the west district enters flood watch, two low-lying roads are marked closed, Shelter A approaches capacity, and a blocked-bridge incident is assigned to a response team. Successful demonstration should show the risk change, a warning-level alert draft, a route avoiding the closed roads, shelter-capacity pressure, and an incident requiring triage.

11. ANALYSIS, COMPARISON, TRADE-OFFS, AND JUSTIFICATION

Design decision	Alternative	Advantages of chosen MVP	Trade-off or limitation
Deterministic mock data	Live sensors and APIs	Reproducible, safe, inexpensive, easy to demonstrate	Does not prove live reliability or prediction accuracy
Rule-based risk score	Trained ML model	Transparent, explainable, no labeled dataset required	Cannot generalize to real hazard conditions without calibration
Local state and localStorage	Database and event streaming	Simple setup and fast prototype iteration	Limited collaboration, durability, auditability,

			and multi-user support
Map component plus fallback	Custom GIS engine	Faster implementation and familiar map interactions	Depends on provider availability and does not replace authoritative GIS operations
Human-approved alerts	Fully automated publication	Safer and more accountable for high-impact decisions	Adds review time and requires trained authorized operators
Responsive web application	Native mobile apps	One codebase, easier distribution, accessible by browser	Offline and device-native features require future work
Fictional region	Real operational jurisdiction	Avoids privacy, security, and miscommunication risks	Limits geographic realism and external validation

The chosen trade-offs are appropriate for an academic MVP because they maximize demonstrable functionality while minimizing safety, privacy, infrastructure, and integration risks. A production system would need formal hazard models, authoritative feeds, security reviews, accessibility testing with users, auditability, service-level objectives, and governance agreements.

12. BROADER CONSIDERATIONS AND SDG RELEVANCE

The project supports the general direction of **SDG 11**, which calls for inclusive, safe, resilient, and sustainable cities and human settlements and identifies disaster-risk reduction as a related topic [3]. It also relates to **SDG 13**, particularly strengthening resilience and adaptive capacity to climate-related hazards and natural disasters. The system's citizen portal, accessible route and shelter information, and multi-agency coordination support resilience conceptually, but the prototype cannot claim measurable SDG impact on its own.

The project also raises important social and ethical considerations. Warnings should be understandable to people with different languages, literacy levels, disabilities, connectivity conditions, and access to transport. Shelter recommendations should consider accessibility, medical needs, vulnerable populations, and capacity rather than distance alone. Citizen reports should minimize personal-data collection and explain how information is used. Internal operational information should not be exposed in the citizen interface.

Because AI-assisted risk assessment may influence high-impact decisions, the design should include transparent contributing signals, uncertainty labels, human review, logging, fallback behaviour, and explicit limitation statements. These choices are consistent with the purpose of the NIST AI Risk Management Framework, which encourages trustworthiness considerations in the design, development, use, and evaluation of AI systems [4].

13. CONCLUSION, LIMITATIONS, AND POSSIBLE IMPROVEMENTS

ResQWatch provides a coherent prototype concept for integrating monitoring, risk assessment, alert management, GIS visualization, evacuation planning, shelter resources, citizen reporting, and response coordination. Its key contribution is not a claim of operational disaster prediction, but the demonstration of how fragmented disaster-management activities can be brought into a shared interface with role-aware workflows and explicit safety boundaries.

The major limitations are the absence of live sensor feeds, validated historical datasets, a production database, real message delivery, official route data, real authentication, real-time multi-user collaboration, and empirical predictive evaluation. The route recommendation and risk scores are simulated and must not be used as official emergency guidance.

Future improvements should include secure role-based authentication, audit logs, PostGIS or an equivalent spatial database, real-time event streaming, validated hazard models, uncertainty calibration, multilingual and accessible alerts, offline-capable citizen reporting, official road and shelter data, SMS/push/email integration, incident deduplication, privacy controls, disaster-recovery planning, and user testing with emergency authorities and community representatives.

14. INDIVIDUAL CONTRIBUTION OF GROUP MEMBERS

Complete this table honestly and obtain agreement from all members.

Member	**Assigned responsibilities**	**Completed deliverables**	**Evidence**	**Approximate contribution**
Ragul SG (192525007)	Requirements, problem formulation, and system scope	Defined the operational problem, functional requirements, assumptions, and safety boundaries	Report Sections 1–3; requirements matrix	14%
Rakkesh Karan R (192524253)	Frontend architecture and application shell	Prepared navigation structure, dashboard layout, reusable UI components, and	Dashboard and application-shell screenshots	14%

		responsive design		
Davis Damien M (192521124)	Risk assessment and algorithms	Developed risk-classification logic, alert-priority logic, pseudocode, and responsible-AI limitations	Algorithm Section and risk-monitoring evidence	14%
Ganta Ram Charan (192525043)	GIS, maps, and evacuation planner	Designed hazard layers, route constraints, evacuation workflow, and map fallback requirements	Architecture diagram; evacuation-planner evidence	14%
Osuru Siddhi Rishwanth (192472148)	Alerts, citizen portal, and communication workflow	Prepared alert-composer flow, citizen-facing guidance, report submission, and communication safeguards	Alert and citizen-portal evidence	15%
Durgam Hithesh (192472167)	Shelter management and response coordination	Defined shelter capacity, resource status, team assignment, and coordination workflows	Shelter and response-coordination evidence	15%
Gandluri Veda Vikas Gowd (192472123)	Testing, validation, and quality assurance	Prepared test cases, validation categories, screenshot index, and evidence boundaries	Test-case table and validation section	14%

The final percentages should follow institutional rules and should not be invented merely to make totals appear equal. Ensure the total is 100% if percentages are required.

15. INDIVIDUAL REFLECTIONS FOR SEVEN STUDENTS

Each reflection below is written as a project-specific draft based on the assigned contribution of one of the seven team members. Students should review the wording, add any personal details, and update the mapped course-outcome number to match the official course document before submission.

15.1 RAGUL SG — 192525007

Role/contribution: Requirements analysis, problem formulation, and system scope.

I contributed to the project by converting the general disaster-management problem into a structured system requirement. I identified the main operational gaps: delayed detection, fragmented agency coordination, limited shelter visibility, inefficient evacuation planning, and weak citizen communication. I also helped define the distinction between simulated MVP functions and future production services. A key design decision was to make the platform a decision-support system rather than claim that it could independently issue certified warnings. This decision was selected because disaster-management software requires human verification, reliable data, and responsible communication.

The main challenge I faced was translating a broad real-world problem into implementable modules without making the project unnecessarily complex. I addressed this by separating the system into monitoring, risk assessment, alerts, GIS, evacuation, shelters, incidents, and response coordination. Through this work, I learned how requirements engineering creates a foundation for architecture, testing, and project evaluation. The project relates especially to SDG 11 because it supports safer and more resilient communities through coordinated risk information. It contributed to the mapped course outcome related to requirements analysis and system design by linking user needs to functional requirements and evidence. My main future improvement would be to validate the requirements with emergency-management professionals and community representatives.

15.2 RAKKESH KARAN R — 192524253

Role/contribution: Frontend architecture, application shell, and responsive interface.

I contributed to the project by designing the frontend structure, navigation shell, dashboard layout, reusable interface components, and responsive page organization. I selected a persistent sidebar and a high-contrast emergency-operations visual style because authorities must be able to move quickly between monitoring, alerts, shelters, incidents, and response teams. The use of reusable cards, badges, tables, drawers, and dialogs improved consistency and reduced repeated interface code.

A major challenge was maintaining readability when the dashboard contained many operational indicators, maps, tables, and charts. I addressed this by introducing a clear visual hierarchy, grouping related functions, using severity labels with icons and text, and considering mobile layouts separately from desktop layouts. I learned how frontend architecture, accessibility, interaction design, and visual communication influence the usability of a safety-critical prototype. The system supports SDG 11 by presenting risk and resilience information in an organized and accessible form. My work contributed to the mapped course outcome related to web development and human-computer interaction through the route structure, reusable components, and responsive layouts. In a future iteration, I would add automated accessibility testing and usability testing with representative authority and citizen users.

15.3 DAVIS DAMIEN M — 192521124

Role/contribution: Risk assessment, alert prioritization, and algorithm documentation.

I contributed to the risk-assessment and algorithmic parts of the project. I prepared the rule-based scoring logic for hazard intensity, exposure, vulnerability, and data quality, and documented pseudocode for risk classification, alert prioritization, route selection, shelter recommendation, and incident prioritization. I selected a transparent rule-based approach because the MVP uses fictional data and does not possess a sufficiently large, labelled historical dataset for a scientifically validated machine-learning model.

The principal challenge was explaining AI/ML functionality without overstating what the prototype could achieve. I addressed this by displaying confidence and contributing signals, labelling the output as simulated decision support, and requiring human verification before alert publication. This assignment improved my understanding of algorithms, data quality, responsible AI, and the difference between a prototype score and a validated predictive model. The project relates to SDG 13 because better risk awareness can support climate adaptation and preparedness, although this prototype does not measure real-world impact. My work contributed to the mapped course outcome related to algorithm design and AI/ML application by connecting input data to explainable decision-support logic. I would improve the system by training and evaluating models on authoritative historical datasets with proper calibration and bias analysis.

15.4 GANTA RAM CHARAN — 192525043

Role/contribution: GIS layers, hazard mapping, and evacuation planning.

I contributed to the GIS and evacuation components by defining hazard zones, shelter markers, emergency facilities, road constraints, route alternatives, and map-layer controls. I chose to compare routes using more than distance alone. Safety score, travel time, accessibility, road closures, and hazard-zone avoidance were included because the shortest route may not be the safest or most suitable option during a disaster.

A major challenge was designing a useful map experience when authoritative geographic and road-condition data were not available. I addressed this by defining fictional Riverdale County data, requiring a map fallback mode, and clearly labelling route recommendations as simulated. I learned how GIS, spatial reasoning, route scoring, and user-centred design interact in emergency applications. The project supports SDG 11 through its focus on safer, more resilient communities and accessible evacuation planning. My work contributed to the mapped course outcome related to GIS, algorithms, or application design by connecting spatial data to evacuation decisions. A future improvement would be integration with authoritative geospatial data, live road closures, traffic information, accessibility data, and field validation.

15.5 OSURU SIDDI RISHWANTH — 192472148

Role/contribution: Alerts, citizen portal, and communication workflow.

I contributed to the alert-composer flow, citizen-facing guidance, incident-reporting interface, and communication safeguards. A key design decision was to require users to select the hazard, severity, affected area, message, expiry, and channels before previewing a demo alert. This was selected because incomplete or ambiguous messages can create confusion, especially during an emergency. I also helped ensure that the citizen portal focused on practical actions such as finding a shelter, checking a route, and reporting an incident.

The main challenge was balancing urgency with accuracy and avoiding language that could be mistaken for an official emergency order. I addressed this by using clear severity labels, visible demo-mode notices, human-review wording, and a separation between authority and citizen information. I learned about communication design, form validation, accessibility, and the importance of plain language. The project supports SDG 11 by improving access to understandable safety information. My contribution relates to the mapped course outcome associated with web applications and user-centred design through the alert and citizen workflows. A future improvement would include multilingual alerts, offline access, official emergency-service integration, and tested SMS or push-notification delivery.

15.6 DURGAM HITHESH — 192472167

Role/contribution: Shelter management and response coordination.

I contributed to the shelter and response-coordination features by defining shelter capacity, occupancy, accessibility, medical support, food, water, sanitation, power, and supply indicators. I also helped structure response-team status, assignments, incident ownership, and activity timelines. I selected a combined capacity-and-resource view because a shelter with available physical space may still be unsuitable if it lacks essential supplies, medical support, or accessibility.

A significant challenge was representing resource changes consistently across the dashboard, shelter detail view, and evacuation workflow. I addressed this by using typed shelter records, occupancy calculations, status badges, update forms, and local state persistence for the MVP. I learned about state management, resource allocation, data modelling, and inter-agency coordination. The project connects to SDG 11 because shelter readiness and coordinated response contribute to safer and more resilient settlements. My work supported the mapped course outcome related to data management and software integration by connecting shelter and response data to operational workflows. Future improvements would include real inventory synchronization, audit logs, secure multi-agency access, and automated resource alerts.

15.7 GANDLURI VEDA VIKAS GOWD — 192472123

Role/contribution: Testing, validation, quality assurance, and evidence preparation.

I contributed by preparing functional test cases, invalid-input scenarios, empty states, map-fallback checks, responsive checks, role-separation checks, and screenshot evidence requirements. I decided to include both successful and failure-oriented tests because a disaster-management application must remain understandable when data is missing, a route is unavailable, a sensor is offline, or a form is incomplete.

A significant challenge was avoiding unsupported claims about performance, predictive accuracy, or operational reliability when the project uses simulated data. I addressed this by separating observed, calculated, simulated, and unavailable results and by marking actual test outcomes for completion after implementation. I learned that validation is not limited to checking whether buttons work; it also requires evidence quality, traceability, accessibility, safety boundaries, and honest reporting. The project supports SDG 11 by encouraging dependable and inclusive community-safety tools. My work contributed to the mapped course outcome related to software testing and quality assurance through the test-case table and evidence framework. Future improvements would include automated end-to-end tests, user testing, performance instrumentation, security testing, and accessibility audits.

16. REFERENCES

[1] United Nations Office for Disaster Risk Reduction (UNDRR). “Definition: Early warning system.” *Sendai Framework Terminology on Disaster Risk Reduction*, 2017. https://www.undrr.org/terminology/early-warning-system

[2] World Meteorological Organization (WMO). “WMO and the Early Warnings for All Initiative.” https://wmo.int/activities/early-warnings-all/wmo-and-early-warnings-all-initiative

[3] United Nations Department of Economic and Social Affairs. “Goal 11: Make cities and human settlements inclusive, safe, resilient and sustainable.” https://sdgs.un.org/goals/goal11

[4] National Institute of Standards and Technology (NIST). “AI Risk Management Framework.” https://www.nist.gov/itl/ai-risk-management-framework

[5] React documentation. “React — The library for web and native user interfaces.” https://react.dev/

[6] Tailwind CSS documentation. “Tailwind CSS.” https://tailwindcss.com/docs

[7] Recharts documentation. “Recharts.” https://recharts.org/

[8] Lucide. “Lucide open-source icon library.” https://lucide.dev/

[9] Google Maps Platform documentation. “Maps JavaScript API.” https://developers.google.com/maps/documentation/javascript